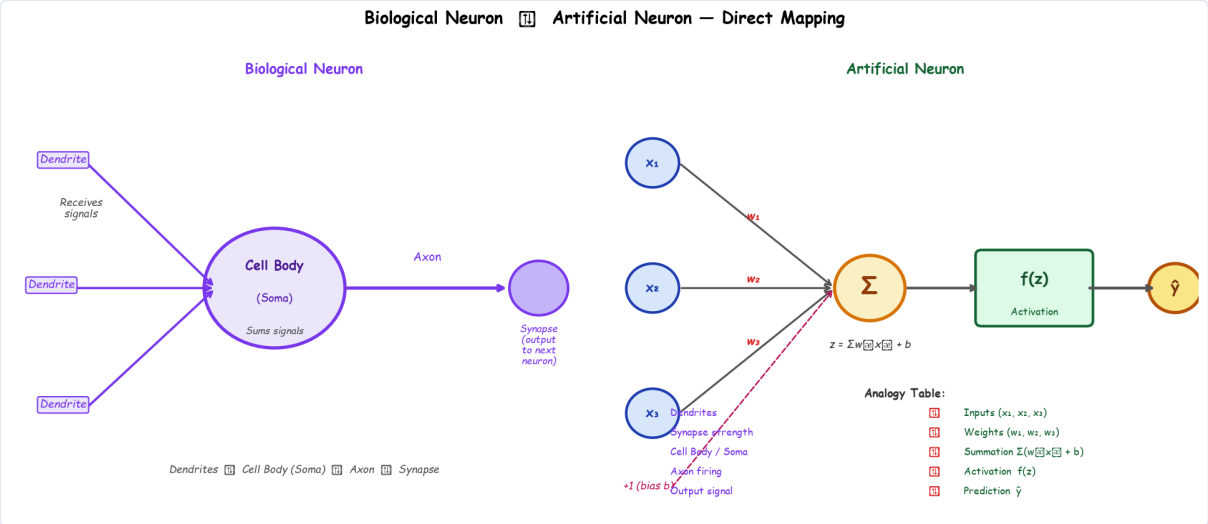
Submodule 36.03: What are Neural Networks?

1. Biological Inspiration: The Human Brain

The foundational architecture of Deep Learning is inspired by the human neurological system. The human brain contains approximately **86 to 100 billion biological neurons**, each interconnected through thousands of synapses, forming a massively parallel information processing network.

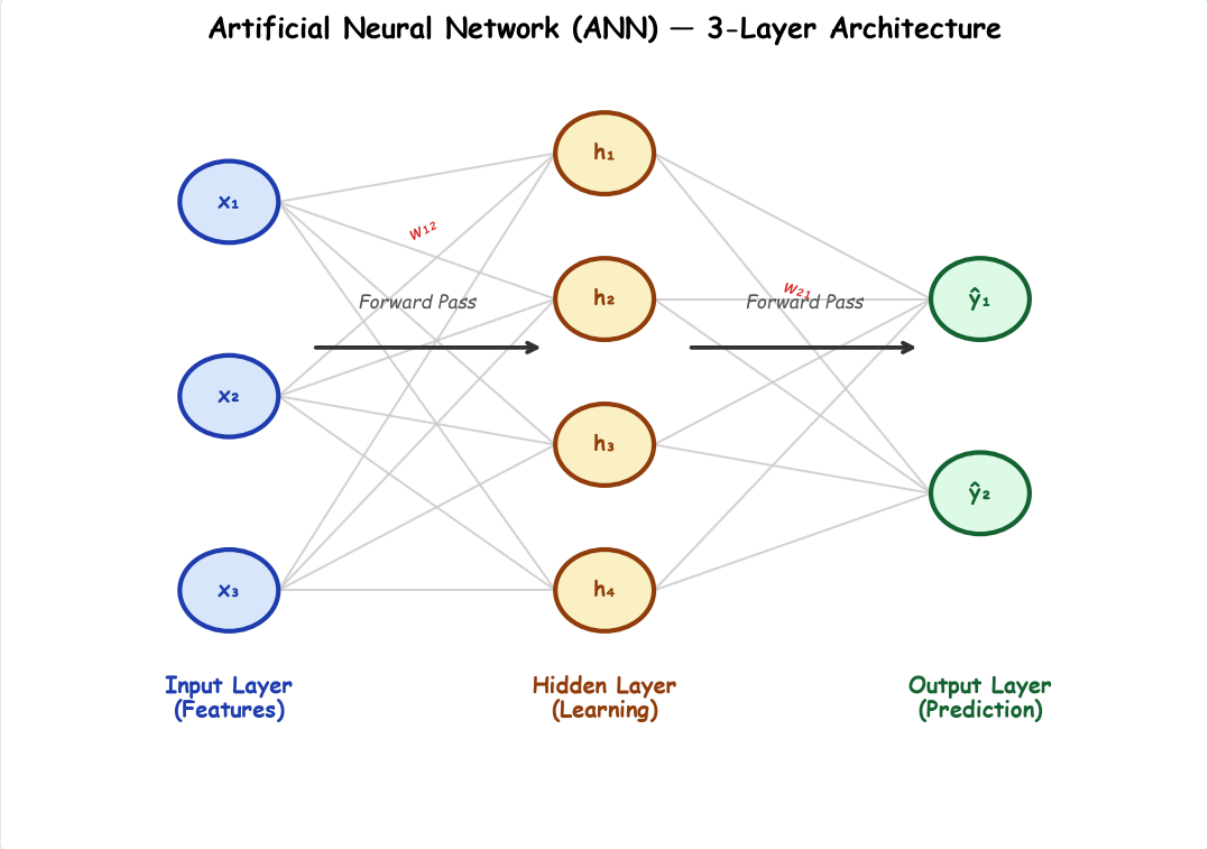
Biological vs Artificial Neuron Mapping:



Biological Component	Artificial Neural Network Equivalent	Computational Role
Dendrites	**Input Features ($x_1, x_2, \ldots, x_n$)**	Receives data inputs from preceding nodes
Synaptic Weights	**Weights ($w_1, w_2, \ldots, w_n$)**	Controls the amplification/attenuation strength of each connection
Cell Body (Soma)	**Summation Node ($\sum w_i x_i + b$)**	Accumulates all weighted inputs plus bias offset
Action Potential Threshold	**Activation Function ($f(z)$)**	Decides whether and with what magnitude the neuron should fire
Axon	**Neuron Output ($\hat{y}$)**	Transmits computed activation signal to downstream layers

2. Artificial Neural Network (ANN) Architecture

An **Artificial Neural Network (ANN)** is organized into discrete sequential groups of artificial neurons called **Layers**:



The Three Fundamental Layer Types:

1. **Input Layer**:
    - The entry point of the network.
    - Contains exactly one node for every input feature (e.g., if predicting house prices using 3 features: `square_feet`, `bedrooms`, `age`, the input layer has 3 nodes).
    - Performs no mathematical transformations; simply passes values forward.
2. **Hidden Layer(s)**:
    - Intermediate layers situated between input and output layers.
    - Perform mathematical dot-products, bias addition, and non-linear activation functions.
    - Extract increasingly abstract feature representations.
    - Networks with multiple hidden layers are called **Deep Neural Networks**.
3. **Output Layer**:
    - The final layer responsible for producing the model's prediction.
    - Node count and activation depend on the problem type:
        - *Binary Classification*: 1 node with **Sigmoid** activation (probability $\in [0, 1]$).

- *Multi-Class Classification*: K nodes with **Softmax** activation (probability distribution summing to 1).
- *Regression*: 1 node with **Linear** (no) activation (continuous real number).